

Stochastic Programming

Chapter 5: Algorithms for Two Stage Stochastic Programs

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Slides based on book: J.R. Birge, F. Louveaux, *Introduction to Stochastic Programming*, Springer, 1997.

Contents

- 1 The Structure of a Stochastic Program
- 2 Benders decomposition for Stochastic Programs
- 3 Example of Bender's decomposition
- 4 Bender's Method Ampliation

Outline

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Matrix structure

Primal and Dual extensive form of a Stochastic Program have next matrix structure:

$$(P) = \min \quad c^\top x + \sum_{i=1}^k p_i q_i y_i$$

$$\begin{aligned} \text{s. to} \quad & \begin{bmatrix} Ax \\ T_1 x \\ \vdots \\ T_k x \end{bmatrix} + \begin{bmatrix} W_1 y_1 \\ \vdots \\ W_k y_k \end{bmatrix} = \begin{bmatrix} b \\ h_1 \\ \vdots \\ h_k \end{bmatrix} \quad \begin{matrix} (u) \\ (v_1) \\ \vdots \\ (v_k) \end{matrix} \\ & x \geq 0 \quad y_i \geq 0 \end{aligned}$$

$$M = \begin{bmatrix} A & & & \\ T_1 & W_1 & & \\ \vdots & & \ddots & \\ T_k & & & W_k \end{bmatrix}$$

$$(D) = \max \quad b^\top u + \sum_{i=1}^k h_i v_i$$

$$\begin{aligned} \text{s. to} \quad & \begin{bmatrix} A^\top u \\ T_1^\top v_1 \\ \vdots \\ T_k^\top v_k \end{bmatrix} + \begin{bmatrix} W_1^\top v_1 \\ \vdots \\ W_k^\top v_k \end{bmatrix} \leq \begin{bmatrix} c \\ p_1 q_1 \\ \vdots \\ p_k q_k \end{bmatrix} \\ & u \in \mathbb{R}, v_i \in \mathbb{R} \end{aligned}$$

$$M^\top = \begin{bmatrix} A^\top & T_1^\top & \cdots & T_k^\top \\ & W_1^\top & & \\ & & \ddots & \\ & & & W_k^\top \end{bmatrix}$$

In general, $W_i = W_j$ if (P) is a fix recourse problem.

- In primal form we have linking variables, so problem can be solved using *Benders Decomposition* and *L-shaped Method*.
- In dual form we have linking constraints, and we'll solve it with *Dantzig-Wolfe Decomposition*

Bibliography about these methods

- Birge & Louveaux, Introduction to SP, Chapter 5, sections 5.1, 5.5 (L-shaped, Dantzig-Wolfe).
- Prekopa, Stoch. Progr., Chapter 12, section 12.4 (L-shaped).
- Kall & Wallace, Stoch. Progr., Chapter 1, section 1.6.4 (General description of dual decomposition); chapter 3, section 3.2 (L-shaped).
- Luenberger, Linear and Nonlinear Programming, section 3.1.0.
- Bradley, Hax, Magnanti, Applied Math Progr., chapter 12 (Dantzig-Wolfe).
- G. Sierksma, Linear and Integer Programming, theory and practice, section 3.6.3 (Benders).

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Benders decomposition

- Method based on L-shaped matrix form.
- For problems which have two groups of variables (x, y) , and variables x “complicate” resolution of problem.
- Benders (1962) designed the method for integer x . In stochastic programming x are first stage variables linking different scenarios.
- The problem to solve is:

$$\begin{aligned}
 (P) = \min \quad & c^\top x + q^\top y \\
 \text{s. to} \quad & Tx + Wy = h \\
 & x \in X \\
 & y \geq 0
 \end{aligned}$$

where $c, x \in \mathbb{R}^{n_1}$
 $q, y \in \mathbb{R}^{n_2}$, $T \in \mathbb{R}^{m \times n_1}$,
 $W \in \mathbb{R}^{m \times n_2}$, and $X \neq \emptyset$
 is any set of
 constraints.

Benders primal and dual subproblems

This method works in the following form

- Fixing $x \in X$, consider the problem

$$\begin{aligned} (Q) = \min \quad & q^\top y \\ \text{s. to} \quad & Wy = h - Tx \\ & y \geq 0 \end{aligned}$$

- Dual of (Q) is

$$\begin{aligned} (Q_D) = \max \quad & u^\top (h - Tx) \\ \text{s. to} \quad & W^\top u \leq q \\ & u \in \mathbb{R}^m \end{aligned}$$

- If (Q_D) has a finite solution then (Q) has a finite solution too; if (Q_D) is unbounded then (Q) is infeasible (so both objectives are $+\infty$).

Benders primal and dual subproblems

- We can write, then

$$\begin{array}{ll} \min & q^\top y \\ \text{s. to} & Wy = h - Tx \\ & y \geq 0 \end{array} \quad \equiv \quad \begin{array}{ll} \max & u^\top (h - Tx) \\ \text{s. to} & W^\top u \leq q \\ & u \in \mathbb{R}^m \end{array}$$

- (P) can be rewritten as:

$$\begin{array}{ll} (P') = \min & c^\top x + \max \{ u^\top (h - Tx) : W^\top u \leq q, \quad u \in \mathbb{R}^m \} \\ \text{s. to} & x \in X \end{array}$$

- Let's consider the feasible set of (Q_D)

$$U = \left\{ u : W^\top u \leq q, \quad u \in \mathbb{R}^m \right\}$$

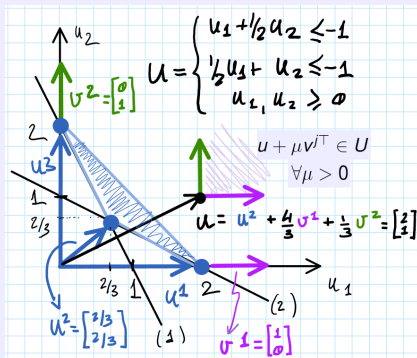
Dual polyhedron

- Consider the set of vertices and extreme rays of U :

$$\begin{cases} u^1, u^2, \dots, u^p & \text{vertices of } U \\ v^1, v^2, \dots, v^q & \text{rays of } U \end{cases}$$

- Hence, all $u \in U$ can be written as

$$\begin{cases} u = \sum_{i=1}^p \lambda_i u^i + \sum_{j=1}^q \mu_j v^j \\ \sum_{i=1}^p \lambda_i = 1 \quad \lambda_i \geq 0 \quad i = 1, \dots, p \quad \mu_j \geq 0 \quad j = 1 \dots q \end{cases}$$



Ensuring (Q) feasibility

- If problem (Q_D) has some extreme ray v^j such that

$$v^{j\top}(h - Tx) > 0, \text{ (ascent dual direction)}$$

then, as $u + \mu v^{j\top} \in U \quad \forall \mu > 0$, we have that the value of the objective function of (Q_D) is $+\infty$.

$$\left(\underbrace{u^\top(h - Tx)}_k + \underbrace{\mu}_{\rightarrow +\infty} \underbrace{v^{j\top}(h - Tx)}_{>0} \right)$$

- Then, (Q_D) will be unbounded and (Q) infeasible. To avoid this, we force:

$$v^{j\top}(h - Tx) \leq 0 \quad j = 1 \dots q.$$

Master Problem

- Hence, problem (P') can be rewritten again as:

$$\begin{aligned}
 (P) \equiv (P') \equiv (P'') \quad \min \quad & c^\top x + \max_{i=1 \dots p} \left\{ u^{i\top} (h - Tx) \right\} \\
 \text{s. to} \quad & v^{j\top} (h - Tx) \leq 0 \quad j = 1 \dots q \\
 & x \in X
 \end{aligned}$$

- We rewrite (P'') adding z , obtaining formulation (BP)

$$\begin{aligned}
 (BP) = \quad \min \quad & z \\
 \text{s. to} \quad & z \geq c^\top x + u^{i\top} (h - Tx) \quad i = 1 \dots p \\
 & v^{j\top} (h - Tx) \leq 0 \quad j = 1 \dots q \\
 & x \in X.
 \end{aligned}$$

Master Problem

- In practice we don't know the p vertices u^i and the q rays v^j , so we generate them on the fly.
- This method considers a relaxation (BP_r) of (BP) with a subset of vertices and extreme rays:

$$\begin{aligned}
 (BP_r) = \quad & \min \quad z \\
 \text{s. to} \quad & z \geq c^\top x + u^{i\top} (h - Tx) \quad i \in I \subseteq \{1 \dots p\} \\
 & v^{j\top} (h - Tx) \leq 0 \quad j \in J \subseteq \{1 \dots q\} \\
 & x \in X
 \end{aligned}$$

- We will denote by (z_r^*, x_r^*) the current solution of (BP_r) , and by (z^*, x^*) the solution of (BP) .

$$(P) = \begin{array}{ll} \min & c^T x + q^T y \\ \text{s. to} & Tx + Wy = h \\ & x \in X \\ & y \geq 0 \end{array}$$

$$(Q) = \begin{array}{ll} \min & q^T y \\ \text{s. to} & Wy = h - Tx \\ & y \geq 0 \end{array}$$

$$(Q_D) = \begin{array}{ll} \max & u^T (h - Tx) \\ \text{s. to} & W^T u \leq q \\ & u \in \mathbb{R}^m \end{array}$$

$$(P') = \begin{array}{ll} \min & c^T x + \max \{ u^T (h - Tx) : W^T u \leq q, u \in \mathbb{R}^m \} \\ \text{s. to} & x \in X \end{array}$$

$$(P'') = \begin{array}{ll} \min & c^T x + \max_{i=1 \dots p} \{ u^i{}^T (h - Tx) \} \\ \text{s. to} & v^j{}^T (h - Tx) \leq 0 \quad j = 1 \dots q \\ & x \in X \end{array}$$

u^i and v^j are fixed parameters for (P'') :
they only depend on the polyhedron $U \Rightarrow$
they don't depend on $x \Rightarrow$ recourse
variables y have disappeared!!!

$$U = \{ u : W^T u \leq q, u \in \mathbb{R}^m \} = \left\{ u : u = \sum_{i=1}^p \lambda_i u^i + \sum_{j=1}^q \mu_j v^j, \sum_{i=1}^p \lambda_i = 1, \lambda_i \geq 0 \quad i = 1, \dots, p, \mu_j \geq 0 \quad j = 1 \dots q \right\}$$

$\{ u^1, u^2, \dots, u^p \}$ vertices of U
 $\{ v^1, v^2, \dots, v^q \}$ rays of U
 $v^j{}^T (h - Tx) < 0 \quad j = 1 \dots q.$
(to prevent taking x that makes (Q) infeasible $\equiv (P)$ infeasible).

$$(BP) = \begin{array}{ll} \min & z \\ \text{s. to} & z \geq c^T x + u^i{}^T (h - Tx) \quad i = 1 \dots p \\ & v^j{}^T (h - Tx) \leq 0 \quad j = 1 \dots q \\ & x \in X. \end{array}$$

To min. w.r.t. z and x
neither u^i nor v^j !!!

(z^*, x^*)

$$(BP_r) = \begin{array}{ll} \min & z \\ \text{s. to} & z \geq c^T x + u^i{}^T (h - Tx) \quad i \in I \subseteq \{1 \dots p\} \\ & v^j{}^T (h - Tx) \leq 0 \quad j \in J \subseteq \{1 \dots q\} \\ & x \in X \end{array}$$

(z_r^*, x_r^*)

Benders algorithm

0. Initializations

1. Solve $(BPr) \rightarrow z_r^*, x_r^*$

2. Solve (Qd) fixing $x = x_r^* \rightarrow (u^*, z^*)$

- a. if (Qd) has a solution compare z_r^* with $z^* = cx_r^* + u^*(h - Tx_r^*)$
 if $z_r^* = z^*$ then
 STOP: optimal solution found
 else

add optimality cut $z \geq cx + u^*(h - Tx)$ to (BPr)

- b. if (Qd) is unbounded then

add feasibility cut $v^*(h - Tx) \leq 0$ to (BPr)
 (and possibly add optimality cut)

3. go to step 1.

Algorithm

Initially

- $I = \emptyset$
- $J = \emptyset$

1 Solve (BP_r) to obtain z_r^* and x_r^* .

▶ $(z_r^* = -\infty \quad \text{any } x_r^* \in X)$

2 Solve (Q_D) using $x = x_r^*$. We can be in two cases:

- ▶ (Q_D) has a finite solution in some vertex u^{i_0} .
- ▶ (Q_D) is unbounded along some extreme ray v^{j_0} .

3 If we haven't found that $x^* = x_r^*$, $z^* = z_r^*$ in the previous step, we go back to step 1.

Case finite solution of (Q_D)

- Suppose (Q_D) has a finite solution in some vertex u^{i_0} . By the equivalence between (P') and (BP) we have that

$$z^* \leq c^\top x_r^* + u^{i_0 \top} (h - T x_r^*)$$

Also, since (BP_r) is a relaxation of (BP) , $z_r^* \leq z^*$

$$\text{Then, } z_r^* \leq z^* \leq c^\top x_r^* + u^{i_0 \top} (h - T x_r^*)$$

- Therefore, if $z_r^* = c^\top x_r^* + u^{i_0 \top} (h - T x_r^*)$ after solving (Q_D) then $z_r^* = z^*$, we have found the optimal solution. **End.**
- Otherwise, this solution (z_r^*, x_r^*) violates the constraint

$$z \geq c^\top x + u^{i_0 \top} (h - T x)$$

of (BP) . This optimality cut is added to (BP_r) : $I \leftarrow I \cup \{i_0\}$

Case unbounded solution of (Q_D)

- Suppose (Q_D) has an unbounded solution along some direction v^{j_0} . This direction is given, for instance, by simplex method computed as

$$v^{j_0 \top} = \left(\begin{array}{c|c} - \underbrace{B^{-1}N_p}_{\text{Basic variables movement}} & \underbrace{0 \dots 1 \dots 0}_{\text{Non basic variables movement}} \end{array} \right)$$

where u^{j_0} is the current point, v^{j_0} the current movement direction, and p the non basic variable that enters the basis.

- In this case, constraint $v^{j_0 \top}(h - Tx) \leq 0$ is violated and it has to be added to (BP_r) .
- Also, as we have found a vertex u^{j_0} , if (z_r^*, x_r^*) violates the constraint $z \geq c^\top x + u^{j_0 \top}(h - Tx)$, we can add it. Therefore, we update $J \leftarrow J \cup \{j_0\}$ and, optionally, $I \leftarrow I \cup \{i_0\}$.

Convergence of Benders decomposition

- The convergence of this algorithm is guaranteed, since the number of constraints is finite.
- In total, we can add up to $p + q$ constraints to (BP_r) , adding one or two in each iteration. Hence, in a maximum of $p + q$ iterations, we finalize.

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Example

- An example using integer variables x and continuous y can be found in G. Sierksma, pages 299-303.
- A stochastic example is the news vendor problem with two scenarios:

$$\begin{array}{ll}
 \min & c^\top x - \sum_{i=1}^2 p_i(qy_i + rw_i) \\
 \text{s. to} & 100 \leq x \leq u \quad \rightarrow \\
 & y_i \leq \xi_i \\
 & y_i + w_i \leq x \\
 & y_i, r_i \geq 0
 \end{array}
 \qquad
 \begin{array}{ll}
 \min & c^\top x - 1/2qy_1 - 1/2rw_1 - 1/2qy_2 - 1/2rw_2 \\
 \text{s. to} & y_1 + s_1 = \xi_1 \\
 & -x + y_1 + w_1 + s_2 = 0 \\
 & y_2 + s_3 = \xi_2 \\
 & -x + y_2 + w_2 + s_4 = 0 \\
 & y_i, w_i, s_j \geq 0 \\
 & 100 \leq x \leq 200
 \end{array}$$

$$\text{Given } \xi_1 = 100 \quad \xi_2 = 200 \quad u = 200 \quad c = 8 \quad q = 10 \quad r = 5$$

Example

- In matrix form, taking the notation of Bender's decomposition:

$$c = \begin{matrix} x \\ [8] \end{matrix} \quad q = \begin{matrix} y_1 & w_1 & s_1 & s_2 & y_2 & w_2 & s_3 & s_4 \\ [-5 & -2.5 & 0 & 0 & -5 & -2.5 & 0 & 0] \end{matrix}$$

$$T = \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \end{bmatrix} \quad h = \begin{bmatrix} \xi_1 \\ 0 \\ \xi_2 \\ 0 \end{bmatrix}$$

$$W = \left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & & & & \\ 1 & 1 & 0 & 1 & & & & \\ \hline & & & & 1 & 0 & 1 & 0 \\ & & & & 1 & 1 & 0 & 1 \end{array} \right] = \left[\begin{array}{c|c} W' & \\ \hline & W' \end{array} \right]$$

Example

$$\mathbf{x} = [x] \quad \mathbf{y} = [y_1 \quad w_1 \quad s_1 \quad s_2 \quad y_2 \quad w_2 \quad s_3 \quad s_4]$$

$$x \in X \equiv \{x : 100 \leq x \leq 200\}$$

- Solution of this problem, applying simplex with all variables, is:

$$x^* = 100 \quad cost^* = -200$$

$$w_1^* = 0 \quad y_1^* = 100$$

$$w_2^* = 0 \quad y_2^* = 100$$

Example

- Applying Bender's we have:

$$(Q) = \min \begin{bmatrix} y_1 & w_1 & s_1 & s_2 & y_2 & w_2 & s_3 & s_4 \end{bmatrix} \cdot \mathbf{y}$$

$$\text{s. to } \left[\begin{array}{c|c} W' & \\ \hline & W' \end{array} \right] \mathbf{y} = \mathbf{h} - \mathbf{T}x = \begin{bmatrix} \xi_1 \\ 0 \\ \xi_2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \end{bmatrix} x = \begin{bmatrix} \xi_1 \\ x \\ \xi_2 \\ x \end{bmatrix}$$

$$y_i, w_i, s_j \geq 0$$

$$(Q_D) = \max \quad u_1 \xi_1 \quad + \quad u_2 x \quad + \quad u_3 \xi_2 \quad + \quad u_4 x$$

$$\text{s. to } \left[\begin{array}{cc|cc} 1 & 1 & & \\ 0 & 1 & & \\ 1 & 0 & & \\ 0 & 1 & & \\ \hline & & 1 & 1 \\ & & 0 & 1 \\ & & 1 & 0 \\ & & 0 & 1 \end{array} \right] \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \leq \begin{bmatrix} -5 \\ -2.5 \\ 0 \\ 0 \\ \hline -5 \\ -2.5 \\ 0 \\ 0 \end{bmatrix}$$

Example

$$U = \{u \in \mathbb{R}^m : W^T u \leq q\}$$

$$(BP_r) = \min \quad z$$

$$\text{s. to} \quad z \geq 8x + (u_1^i, u_2^i, u_3^i, u_4^i) \begin{pmatrix} 100 \\ x \\ 200 \\ x \end{pmatrix} \quad i \in I$$

$$(v_1^j, v_2^j, v_3^j, v_4^j) \begin{pmatrix} 100 \\ x \\ 200 \\ x \end{pmatrix} \leq 0 \quad j \in J$$

$$100 \leq x \leq 200$$

Example

- Apply the algorithm:

Initially $I = \emptyset$ $J = \emptyset$

- Solve (BP_r) :

- $z_r^* = -\infty$ $z_r^* = -\infty$
- Any x such that $100 \leq x \leq 200$, for instance $x_r^* = 200$.

- Solve (Q_D) taking $x_r^* = 200$:

$$\begin{array}{ll} \max & 100u_1 + 200u_2 + 200u_3 + 200u_4 \\ \text{s. to} & W^\top u \leq q \end{array} \quad \rightarrow \quad \begin{array}{l} u^{1\top} = (-2.5 \quad -2.5 \quad -1.0365 \quad -3.9635) \\ f^* = -1750 \end{array}$$

$$z_r^* = -\infty < c^\top x_r^* + u^{1\top} (h - T x_r^*)$$

- We add a new constraint to (BP_r) :

$$z \geq c^\top x + u^{1\top} (h - T x) = 8x + (-2.5)100 - 2.5x + 200(-1.0365) + x(-3.96359)$$

$$z \geq 1.536534x - 457.3069$$

- Go to step 1.

Example

1 Solve (BP_r)

$$\begin{array}{ll} \min & z \\ \text{s. to} & z \geq -457.3 + 1.5365x \longrightarrow \\ & 100 \leq x \leq 200 \end{array}$$

Solution

$$x_r^* = 100$$

$$z_r^* = -303.657$$

2 Solve (Q_D) taking $x = x_r^* = 100$

$$\begin{array}{ll} \max & 100u_1 + 100u_2 + 200u_3 + 100u_4 \longrightarrow u^{2\top} = (-0.9038 \quad -4.096 \quad 0 \quad -5) \\ \text{s. to} & W^\top u \leq q \quad f^* = -1000 \end{array}$$

$$z_r^* = -303.657 < c^\top x_r^+ + u^{2\top} (h - T x_r^*)$$

► We add a new constraint to (BP_r) :

$$z \geq 8x + (-0.9038)100 - 4.096x + 5x$$

$$z \geq -1.096152x - 90.38$$

3 Go to step 1.

Example

1 Solve (BP_r)

$$\begin{array}{ll} \min & z \\ \text{s. to} & z \geq -457.3 + 1.5365x \\ & z \geq -1.096152x - 90.38 \\ & 100 \leq x \leq 200 \end{array} \longrightarrow$$

Solution

$$x_r^* = 139.375$$

$$z_r^* = -243.75$$

2 Solve (Q_D) taking $x = x_r^* = 139.375$, and continue.

- What this method does through the planes $z \geq a + bx$ is to approximate $cx + Q(x)$.
- Let's see it:

$Q(x) = 1/2Q(x, \xi_1) + 1/2Q(x, \xi_2)$. In this case, where

$$Q(x, \xi) = \begin{cases} -qx & \text{if } x \leq \xi \\ -q\xi - r(x - \xi) & \text{if } x \geq \xi \end{cases}$$

$$q = 10$$

$$r = 5$$

$$c = 8$$

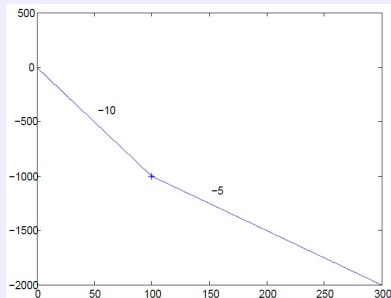
$$\xi_1 = 100$$

$$\xi_2 = 200$$

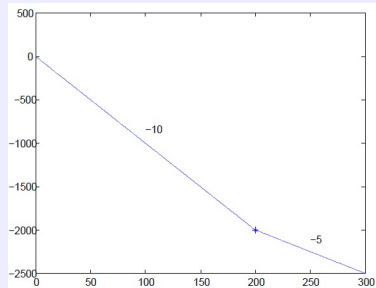
Example

Then:

$$Q(x, \xi_1 = 100)$$



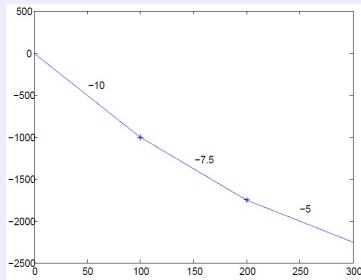
$$Q(x, \xi_2 = 200)$$



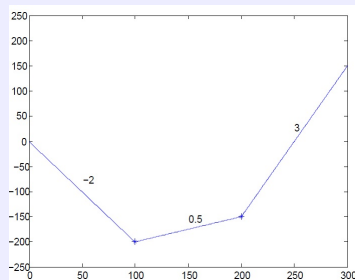
Example

Therefore:

$$Q(x)$$



$$8x + Q(x)$$



Example

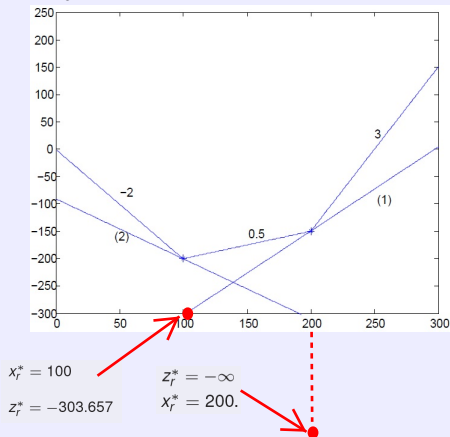
The minimum for $100 \leq x \leq 200$ is clearly $x = 100$, with cost = 200.

The two planes we have generated are:

① $z \geq 1.536534 \cdot x - 457.3069$

② $z \geq -1.096152 \cdot x - 90.38$

These planes go approaching $cx + Q(x)$



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Bender's Modification

In practice, is easier to solve (Q) than its dual (Q_D)

$$\begin{aligned} (Q) \quad & \min \quad q^\top y \\ & \text{s. to} \quad Wy = h - Tx \\ & \quad y \geq 0 \end{aligned}$$

$$\begin{aligned} (Q_D) \quad & \max \quad u^\top (h - Tx) \\ & \text{s. to} \quad W^\top u \leq q \\ & \quad u \in \mathbb{R}^m \end{aligned}$$

- If (Q_D) has solution also has (Q) , and the u^{i_0} solution of (Q_D) can be obtained through the solution of (Q) : u^{i_0} are the dual variables at the optimum of (Q) .
- Problem comes when (Q_D) is unbounded ($u^{i_0} + \lambda v^{j_0}$ is the solution). If we solve (Q) we'll find a infeasible problem: we have to find u^{i_0} and v^{j_0} from (Q) .
- Benders, in his paper of 1962 proposes a way to find u^{i_0} and v^{j_0} in the case of (Q_D) , computing (Q) . The L-Shaped method of Van Slyke and Wets proposes another way (is the principal difference between original Benders and L-Shaped).

Formulation of (Q) that avoids infeasibility

$$(Q) \quad \begin{array}{ll} \min & q^\top y \\ \text{s. to} & Wy = h - Tx \\ & y \geq 0 \end{array} \Leftrightarrow$$

Taking $u = u^+ - u^-$, its duals are:

$$(Q_D) \quad \begin{array}{ll} \max & u^\top (h - Tx) \\ \text{s. to} & W^\top u \leq q \\ & u \in \mathbb{R}^m \end{array} \Leftrightarrow$$

$$(Q) \quad \begin{array}{ll} \min & q^\top y \\ \text{s. to} & Wy \geq h - Tx \\ & -Wy \geq -(h - Tx) \\ & y \geq 0 \end{array} \quad \begin{array}{l} \text{multipliers} \\ (u^+) \\ (u^-) \end{array}$$

$$(Q_D) \quad \begin{array}{ll} \max & u^+(h - Tx) - u^-(h - Tx) \\ \text{s. to} & W^\top u^+ - W^\top u^- \leq q \\ & u^+ \quad u^- \geq 0 \end{array}$$

To avoid (Q_D) undounded ($\Leftrightarrow (Q)$ infeasible), we impose:

$$\begin{aligned} e^\top u^+ &= \sum_{i=1}^m u_i^+ \leq M \\ e^\top u^- &= \sum_{i=1}^m u_i^- \leq M \end{aligned}$$

for a large M .

Formulation of (Q) that avoids infeasibility

Then, dual and primal problem are:

$$\begin{array}{ll}
 (Q_D^M) = \max & u^+(h - Tx) - u^-(h - Tx) \\
 \text{s. to} & W^T u^+ - W^T u^- \leq q \\
 & e^T u^+ \leq M \\
 & e^T u^- \leq M \\
 & u^+ - u^- \geq 0
 \end{array}
 \quad
 \begin{array}{l}
 y \in \mathbb{R}^n \\
 w^+ \in \mathbb{R} \\
 w^- \in \mathbb{R}
 \end{array}
 \quad
 \begin{array}{ll}
 (Q^M) = \min & q^T y + Mw^+ + Mw^- \\
 \text{s. to} & Wy + w^+ e \geq h - Tx \\
 & Wy + w^- e \geq -(h - Tx) \\
 & y \quad w^+ \quad w^- \geq 0
 \end{array}
 \quad
 \begin{array}{l}
 y \in \mathbb{R}^n \\
 u^+ \\
 u^-
 \end{array}$$

Solve (Q^M) :

- If $w^+ = w^- = 0$: (Q^M) is feasible, and we obtain u through its dual variables $u = u^+ - u^-$.
- If $w^+ + w^- > 0$: (Q^M) is infeasible. From the optimal solution of (Q^M) we can obtain the vertex and extreme ray $u^{j_0} + \lambda v^{j_0}$ from the dual variables of the problem

$$u^M = B^{-1T} \cdot c_B$$

c_B includes costs q and M , because some w^+ or $w^- > 0$ and then enters the basis.

Then, we can write $u^M = u^1 + M \cdot u^2$

and from this decomposition it is obtained u^{j_0} and v^{j_0} (see Benders paper for details).